

ML-1

Data Column $\begin{cases} \rightarrow \text{Input} \\ \rightarrow \text{Output} \end{cases}$

Input $\rightarrow$ Feature Column / Attribute Column

Output $\rightarrow$ label / Target Column

I/O $\rightarrow$ labeled Dataset

O/P $\rightarrow$ unlabeled "

3 way of learning $\begin{cases} \text{Statistical ML} \\ \text{DL} \\ \text{NLP} \end{cases}$

Traditional Programming:

Input / Parameter $\rightarrow$ Algo $\rightarrow$ Output

It is problem specific.

ML:

Input Dataset $\rightarrow$ ML Model
Algo $\rightarrow$ Pattern, Algo, Rule

It finds out pattern/design algo to solve prob.

epoch $\rightarrow$ One particular Time period

Prediction $\rightarrow$ For uncertain (correct / wrong) event

Learning:

① Supervised Learning: labeled dataset needed

② Unsupervised " : no labeled " "

③ Reinforcement " :

Task for this Course:

1. Train an ML Model

2. Evaluate / Validate (Cross Validation)

3. Testing

4. Paper Publication (at least Conference one)

ML-2

SL

USL

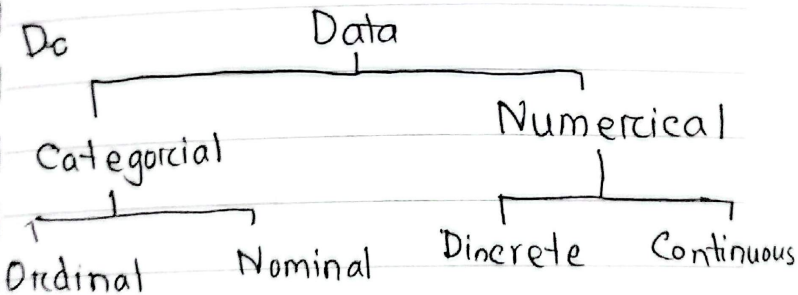
RL

1. Labeled
dataset

1. Unlabeled

1. No dataset

Dimension $\rightarrow$ ndim / rank (no. of axis)
req



Feature

data point $\wedge$ Vector / Tensor /
ds row $\rightarrow$ Data Sample / Data Example

- graphical representation $\rightarrow$ Visualiz-
ation / Exploratory Data Analysis

Shape: no of elements in each dimension

tensor: a SDS used to store way of number

shape = (row, column)

ex: shape = (8, 5, 3)

↳ There r 3 8X5 matrix

f_1	f_2	...	f_n	y	
x_{11}	x_{12}		x_{1n}	y_1	x
.	.		.	.	
.	.		.	.	
x_{m1}	x_{m2}		x_{mn}	y_n	X

Batch of Data Point $\downarrow$ 2D Tensor
mini-Batch

- If parameter reached optimal value, then ML models r properly trained.

SD

ML Stage Life Cycle (MDLC)

1. Frame Data Science / ML prob.
2. Data Collection
3. Data Preprocessing
4. EDA (Exploratory Data Analysis)
5. Feature Selection & Engineering
6. Model Training, Validation / Evaluation, Selection
7. Model Deployment, 8. Testing
9. Optimization & Hyper parameter tuning

If o/p is Continuous $\rightarrow$ Regression

Supervised L

└ Classification Prob
└ Regression "

optimal ~~profit~~ cost parameter $\rightarrow$
Hyper Parameter.

Machine Learning Software
Development Life Cycle.

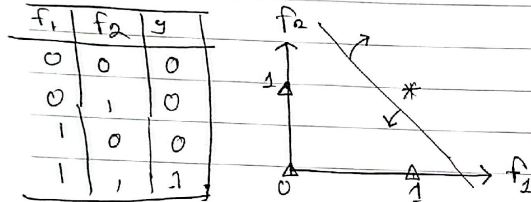
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ML-3

Linear Regression (LR):

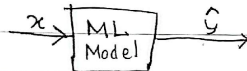
2-input AND Logic



- It's Binary Classification Prob.

Class = 2, Binary " "

3, ternary " "

~~2~~ ≥ 4 multi-class " "Classification: Output $\rightarrow$ discrete,
category, class $y \rightarrow$ Actual O/P $\hat{y} \rightarrow$ Predicted O/P

DATE

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Supervised

- Classification
 - Binary
 - Ternary
 - multi-class

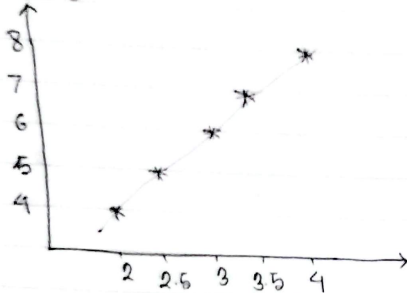
Regression

- Linear
 - Simple
 - multiple LR
- Polynomial
- Logistic
- Ridge Lasso

Build a regression model to predict the o/p $\hat{y}$.

I.D.S	(x) CGPA	Salary Offered (y)
	2	4
	2.5	5
	3	6
	3.5	7
	4	8

y (Salary)



$x \uparrow, y \uparrow$: Positive Linear Relationship

$x \uparrow, y \downarrow$: Negative " "

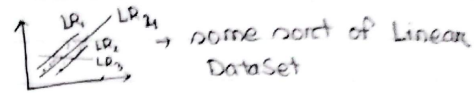
Let, $\hat{y} = \omega x + b$ | b = intersect of y
 ω = slope

$$\omega = \frac{y_2 - y_1}{x_2 - x_1}$$

here, $\omega = 2, b = 0$

$$\hat{y} = 2x + 0$$

$$\hat{y} = 2x$$



LR_4 : best fitted LR Line

error in the amt of deviation
 generated by ML from actual o/p.

(x) CGPA	Salary Offered (y)	Predicted Salary ($\hat{y}$)	Error (e)
x_1	y_1	$\hat{y}_1$	$y_1 - \hat{y}_1$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_i	y_i	$\hat{y}_i$	$y_i - \hat{y}_i$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_m	y_m	$\hat{y}_m$	$y_m - \hat{y}_m$

$$\text{Total error/loss} = \sum_{i=1}^m e_i = \sum_{i=1}^m L_i$$

Cost, C.

$$\text{or, } \sum_{i=1}^m L_i = \sum_{i=1}^m (y_i - \hat{y}_i)$$

Best fit & R. line : cost func. value is minimum

$$C(w, b) = \sum_{i=0}^m |y_i - \hat{y}_i| ; \text{Absolute Cost func}$$

$$\text{Mean ACF (MAE)} = \frac{1}{m} \sum_{i=0}^m |y_i - \hat{y}_i|$$

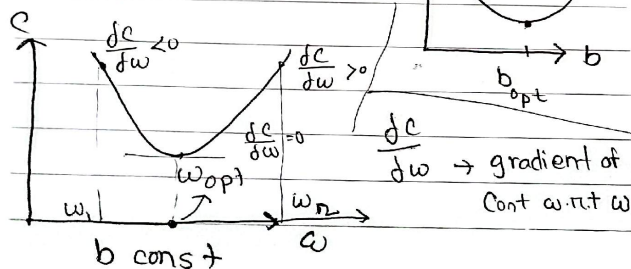
(MAR)

Error / Cost / Residual

$$\text{MSE} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\text{Residual Squared Sum} = \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\hat{y} = \underbrace{w}_{\text{const}} x + \underbrace{b}_{\text{variable}}$$



while $(\frac{dC}{dw} \neq 0)$ for $(i = 1 \text{ to } 10000)$

$$w = \text{rand}()$$

if $\frac{dC}{dw} < 0$: (Gradient, $w + \frac{dC}{dw}$ until w is $\frac{dC}{dw}$)

$$w = w + \eta \frac{dC}{dw}$$

if $\frac{dC}{dw} > 0$:

$$w = w - \eta \frac{dC}{dw}$$

y

$$\text{if } \frac{dC}{dw} = 0 : \text{break}$$

- Training quality based on Hyperparameter

- Hyperparameter Tuning

- η = Learning rate

Gradient Decent Algo

gradient descent (x, y, n)

1. $w = \text{rand}()$

2. $b = \text{rand}()$

3. for $k = 1$ to 10000

3.i) Compute $\frac{\partial C}{\partial w}$

3.ii) Compute $\frac{\partial C}{\partial b}$

3.iii) $w = w - \eta * \frac{\partial C}{\partial w}$

3.iv) $b = b - \eta * \frac{\partial C}{\partial b}$

4. return w, b

i) Batch GD

ii) Min Batch GD

iii) Stochastic GD

ML-4

$$X =$$

x	y	$\hat{y}$
x_1	y_1	$\hat{y}_1$
x_2	y_2	$\hat{y}_2$
$\vdots$		
x_m	y_m	$\hat{y}_m$

$$C = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2 \rightarrow \text{RSS}$$

epoch is a time-period when in which entire dataset is being presented

$$C = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

$$\frac{\partial C}{\partial w} = \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial w} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{m} \sum_{i=1}^m \frac{\partial}{\partial \hat{y}_i} \{ (y_i - \hat{y}_i)^2 \} \cdot \frac{\partial \hat{y}_i}{\partial w}$$

$$= \frac{1}{m} \sum_{i=1}^m 2(y_i - \hat{y}_i) (-1) \cdot \frac{\partial \hat{y}_i}{\partial w} (wx_i + b)$$

$$= -\frac{2}{m} \sum_{i=1}^m (y_i - \hat{y}_i) x_i$$

Similarly, $\frac{\partial c}{\partial b} = -\frac{2}{m} \sum_{i=1}^m (y_i - \hat{y}_i)$

$$\text{as } \frac{\partial \hat{y}_i}{\partial b} = \frac{\partial}{\partial b} (\omega x_i + b) = 1.$$

Let, $\omega_sum = \sum_{i=1}^m (y_i - \hat{y}_i) x_i = 0$

$$b_sum = \sum_{i=1}^m (y_i - \hat{y}_i) = 0$$

batch_GD(X, y)

Algo:

1. $\omega_sum = 0$, $b_sum = 0$

2. for $i = 1$ to m

2.1 $\omega_sum += (y_i - \hat{y}_i) * x_i$

2.2 $b_sum += (y_i - \hat{y}_i)$

3. $\frac{\partial c}{\partial \omega} = -\frac{2}{m} * \omega_sum$

4. $\frac{\partial c}{\partial b} = -\frac{2}{m} * b_sum$

5. return $\frac{\partial c}{\partial \omega}$, $\frac{\partial c}{\partial b}$

$$\left. \begin{aligned} \hat{y}_1 &= \omega x_1 + b \\ \hat{y}_2 &= \omega x_2 + b \\ \hat{y}_3 &= \omega x_3 + b \\ &\vdots \\ \hat{y}_m &= \omega x_m + b \end{aligned} \right\} y_i \hat{y} = \omega X + b$$

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix}_{m \times 1}$$

$\omega, b \rightarrow$ scalar term

$E = y - \hat{y}$, matrix: $E_{m \times 1}$

here, ω_sum is a scalar value

but $E_{m \times 1}, X_{m \times 1} \rightarrow$ can't be multiplied to produce scalar value ($\odot_{1 \times 1}$)

So, we need to transpose one.

That's why,

$$\frac{\partial c}{\partial \omega} = -\frac{2}{m} X^T E$$

$$= -\frac{2}{m} \begin{bmatrix} x_1 & x_2 & \dots & x_m \end{bmatrix}_{1 \times m} \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_m - \hat{y}_m \end{bmatrix}_{m \times 1}$$

$$= -\frac{2}{m} \left[(y_1 - \hat{y}_1) x_1 + (y_2 - \hat{y}_2) x_2 + \dots + (y_m - \hat{y}_m) x_m \right]$$

Disadv of Batch-GD / Gradient Descent

1. High time consuming approach

$O(nm)$ $O(NM)$

2. but given best ans.

- Sample size 30 $\rightarrow$ mainly provides very true closer to true value.

- to avoid this, we can consider mini-batch, instead of batch (full dataset)

that gives:

mini-batch-GD (x, y)

1.

2. for $i = 1$ to k

$$\frac{\partial c}{\partial w} = -\frac{2}{k} * w_{sum}, \quad \frac{\partial c}{\partial b} = -\frac{2}{k} * w_{sum}$$

$$\frac{\partial c}{\partial w} = -\frac{2}{k} X^T E$$

- mini-batch $\hookrightarrow$ convergence

($w \rightarrow w_{opt}$ + iterations process) faster than batch-GD.

$K \downarrow$ Acc $\downarrow$ Time $\downarrow$.

- Stochastic GD to $k=1$.

SGD (x, y, η)

1. $w = \text{rand}()$

2. $b = \text{rand}()$

3. for $i = 1$ to 10^5 :

for $i = 1$ to m :

$$\frac{\partial c}{\partial w} = -\frac{2}{k} * (y_i - \hat{y}_i) x_i$$

$$\frac{\partial c}{\partial b} = -\frac{2}{k} * (y_i - \hat{y}_i)$$

$$w = w + 2\eta (y_i - \hat{y}_i) x_i$$

$$b = b + 2\eta (y_i - \hat{y}_i)$$

j

এই $10^5 \rightarrow 1000$
max ফিটনেস
closer true
value পাওয়া
যাবে.

Q: Diff betⁿ Batch_GD, mini-GD, SGD

BGD	mini-GD	SGD
W, b conver- gence slower	much faster	faster
higher acc	comparatively lower acc than BGD	lower than all

Simplification of SGD (3)

$$w = w + \eta (y_i - \hat{y}_i) x_i$$

$$b = b + \eta (y_i - \hat{y}_i)$$

H.W Regression Model, $\hat{y} = wx + b$
using Linear Algebra eqⁿ

epoch
BGD $\rightarrow$ matrix use $\odot$
mini-BGD $\rightarrow$ random sample $\rightarrow$ matrix
SGD $\rightarrow$ linear (no matrix, manual
computation)

ML-5

Age data of these students who
has completed Graduation.

Ques: what is the
expected age of
a student at the time
of their graduation

	Age
x_1	24
x_2	25
	26
	23
	24
	22
	27
Sample: x_1	24

Population: the entire
grp that u study

Sample:

Sample size = n

population size = N

expected age

$\rightarrow$ Mean Lverage

$$\text{Sample mean, } \bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$= \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Population mean, } \mu = \frac{x_1 + x_2 + \dots + x_n}{N}$$

$$= \frac{1}{N} \sum_{i=1}^n x_i$$

$$\bar{x} = \frac{24 + 25 + 26 + 23 + 24 + 22 + 27 + 24}{8}$$

$$= 24.375$$

$$\approx 24 \text{ years}$$

$$\text{Variation, } S^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$S^2 = 4 \text{ year}^2$$

$$S = \sqrt{4 \text{ year}^2} = 2 \text{ year}$$

The actual age of it deviates by

$S = 2$ years from the expected

age $\bar{x} = 24$ years.

$$\text{Normal range of ages} = [\bar{x} - S, \bar{x} + S]$$

$$= [24 - 2, 24 + 2]$$

$$= [22, 26]$$

For population: $[\sigma - 1]$

$$[\mu - \sigma, \mu + \sigma]$$

$$\text{maximum tolerance } [\mu - 3\sigma, \mu + 3\sigma]$$